A

Report on

PRACTICAL SECURITY ASSESSMENT (TASK 8)

Submitted In Partial Fulfillment of the Requirements

for the

Cyber Security Internship (Week 3) (VAPT)

Internship Organization

CyArt Tech LLP

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1. INTRODUCTION

1.1 Overview

This report presents the practical implementation of Vulnerability Assessment and Penetration Testing (VAPT). It includes exploitation techniques, web application testing, vulnerability reporting, and post-exploitation analysis conducted in a controlled lab environment.

1.2 Purpose of This Task

The purpose of this task is to develop hands-on skills in identifying, exploiting, and reporting vulnerabilities while following industry-standard methodologies.

1.3 Target Information

- Target Systems: Metasploitable2, DVWA, VulnHub Machines
- Environment: Virtual Lab Setup
- Tools Used: Kali Linux, Metasploit, Burp Suite, SQLMap, Wireshark

1.4 Objectives

- Perform advanced exploit chaining
- Identify web vulnerabilities (OWASP Top 10)
- Gain system access using exploitation tools
- Document findings professionally

2. ADVANCED EXPLOITATION LAB

2.1 Exploit Chain Execution

An exploit chain was performed where a Cross-Site Scripting (XSS) vulnerability was used to gain access, followed by escalation to Remote Code Execution (RCE).

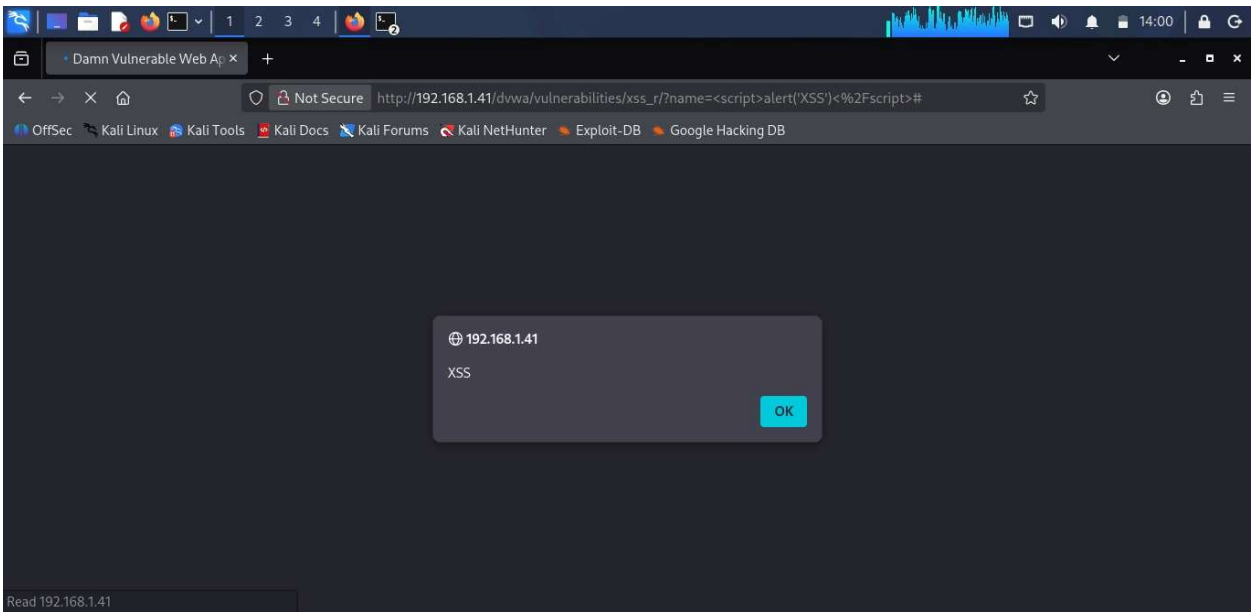

Figure 1: Exploit Chain Execution (XSS to RCE)

2.2 Metasploit Configuration & Execution

Metasploit Framework was configured with:

- Target IP address
- Payload: Meterpreter reverse shell
- Exploit module selection

The exploit was executed successfully, resulting in system access.

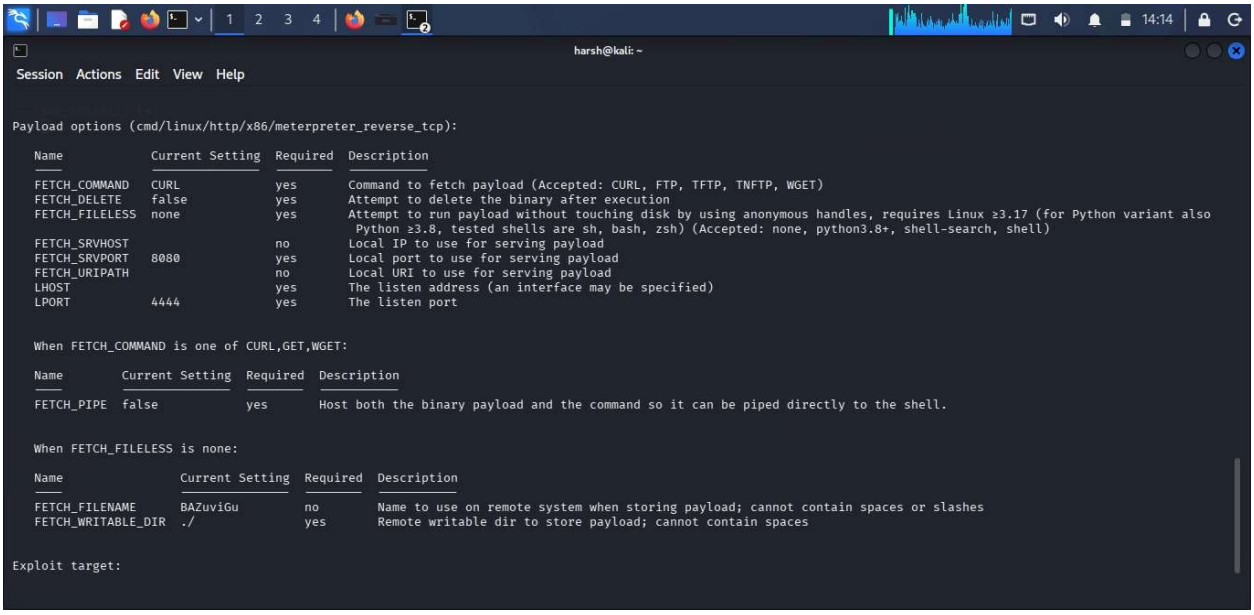

Figure 2: Metasploit Exploit Configuration

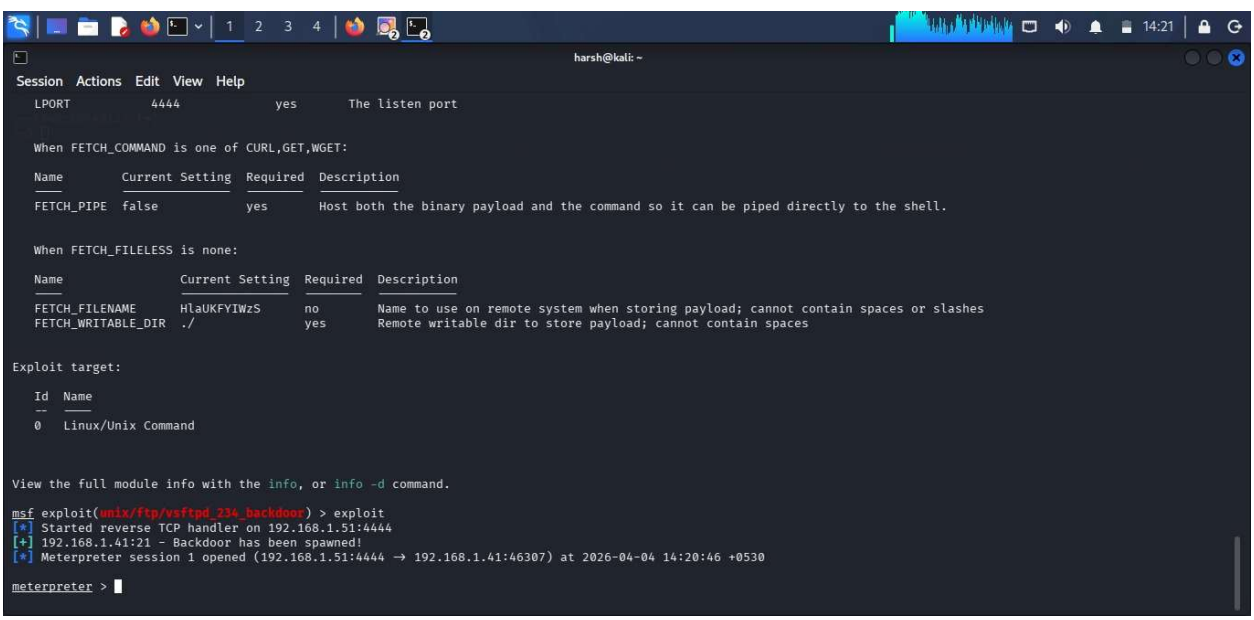

Figure 3: Metasploit Exploit Execution

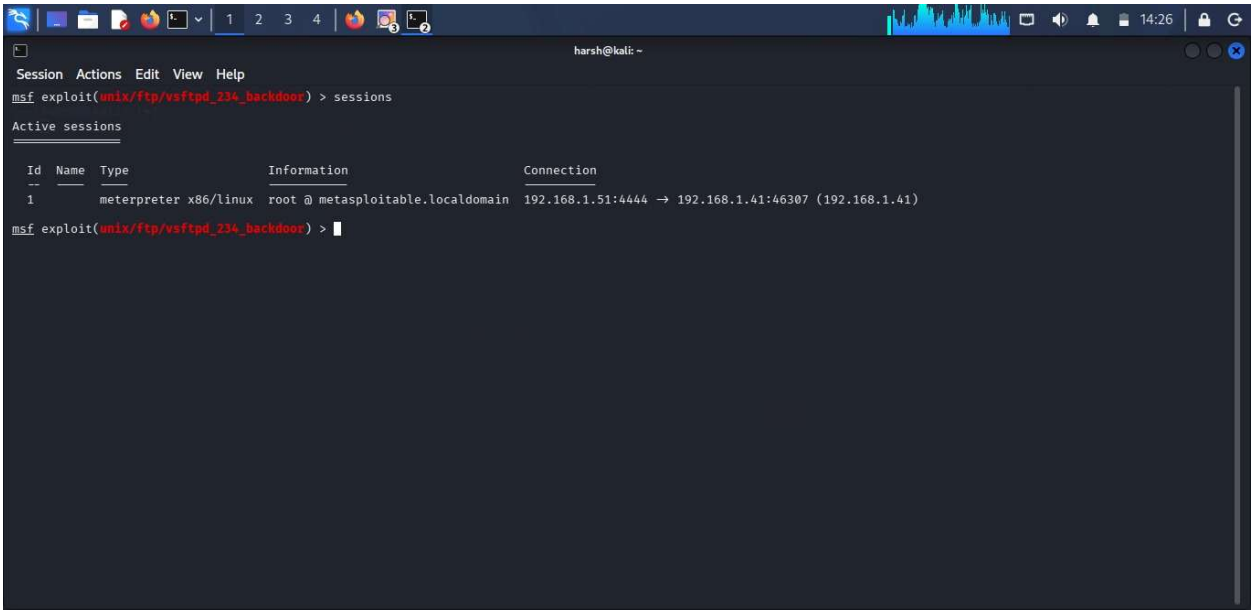

Figure 4: Meterpreter Session Established

2.3 Exploit Customization

Existing exploit code was modified to match the target environment. Parameters such as IP address, payload, and encoding were adjusted.

Before Customization:

- Default exploit code from Exploit-DB
- Fixed payload configuration

After Customization:

- Modified target IP
- Adjusted payload delivery
- Added error handling

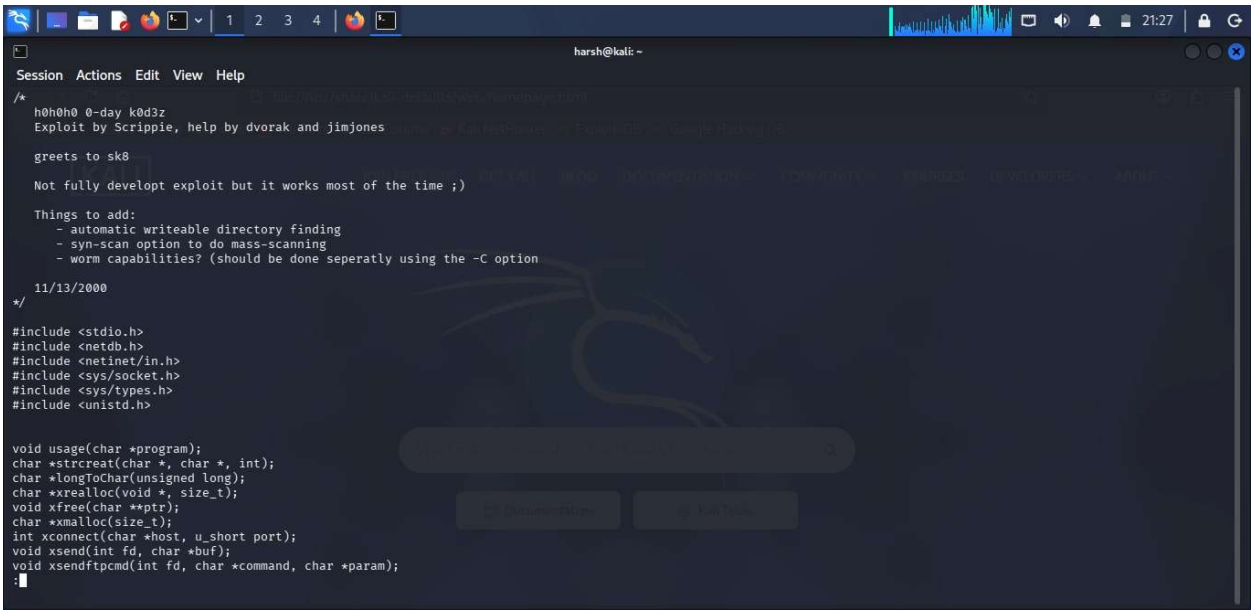

Figure 5: Exploit Code Before Customization

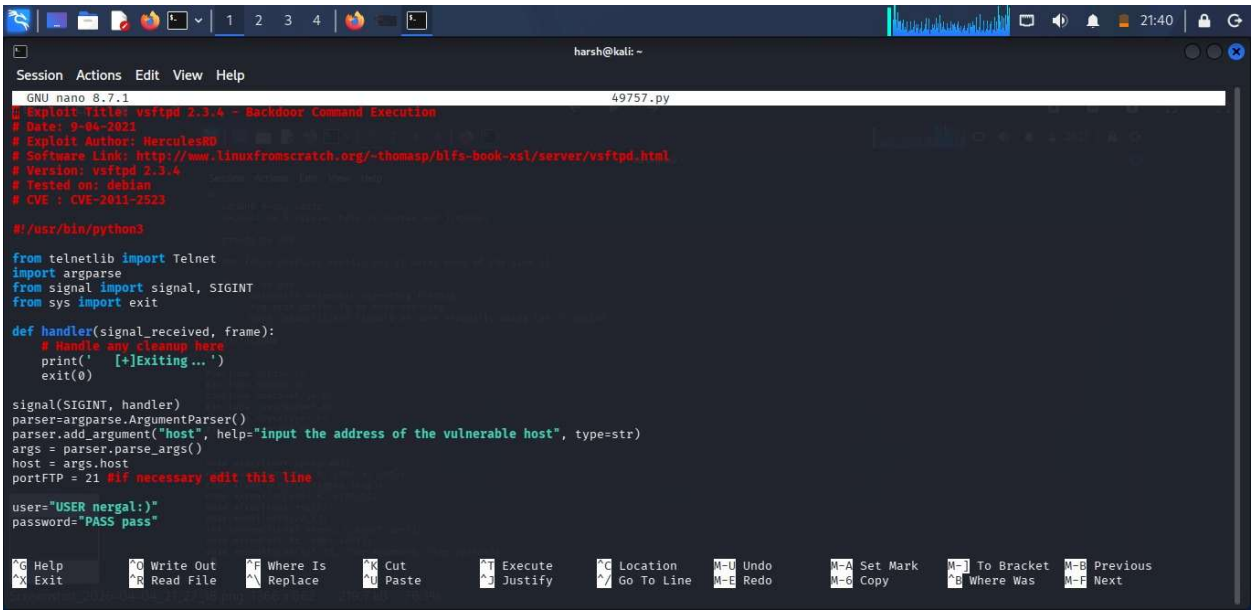

Figure 6: Exploit Code After Customization

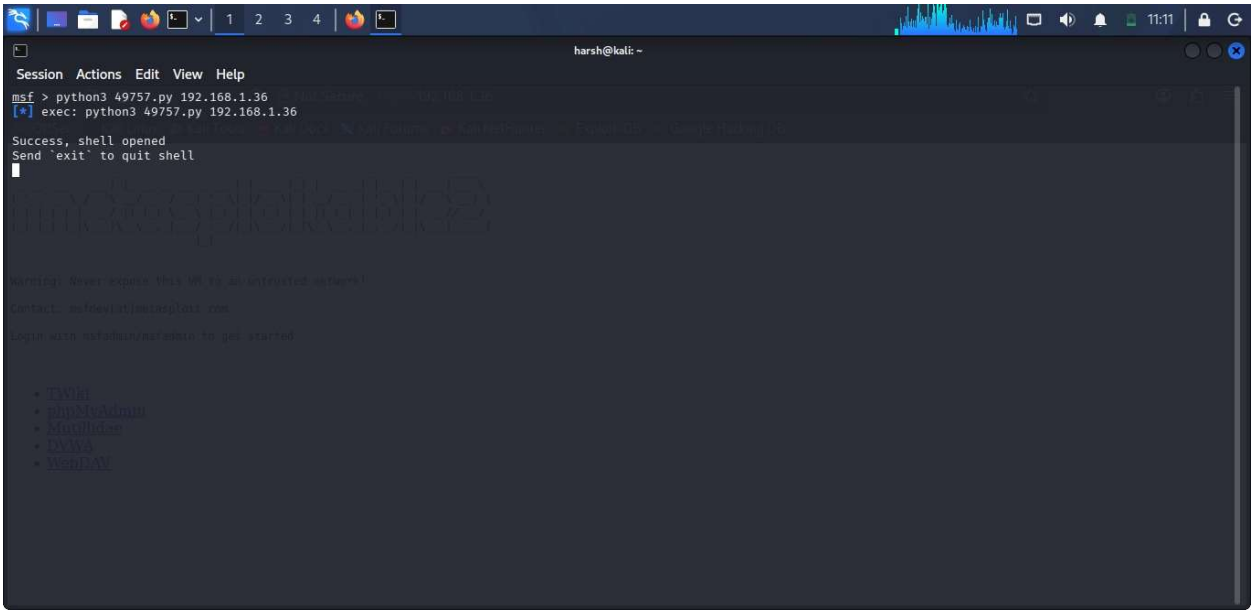

Figure 7: Custom Exploit Execution Result

3. WEB APPLICATION TESTING LAB

3.1 SQL Injection Testing

SQL Injection vulnerabilities were tested using manual payloads and automated tools like sqlmap.

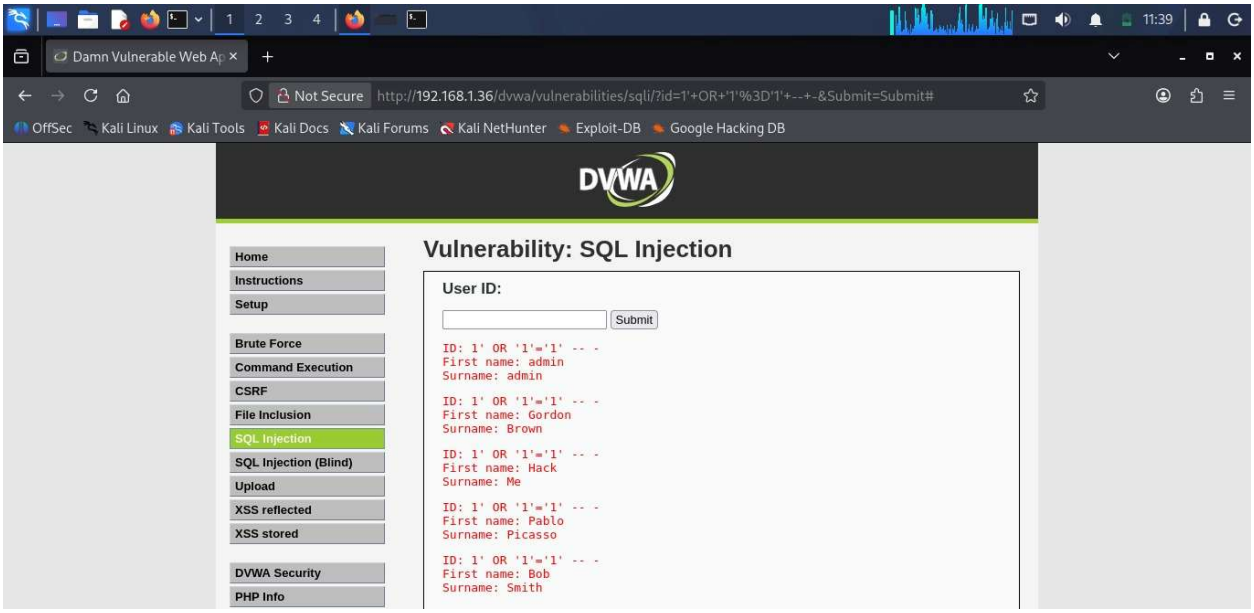

Figure 8: SQL Injection Payload Testing

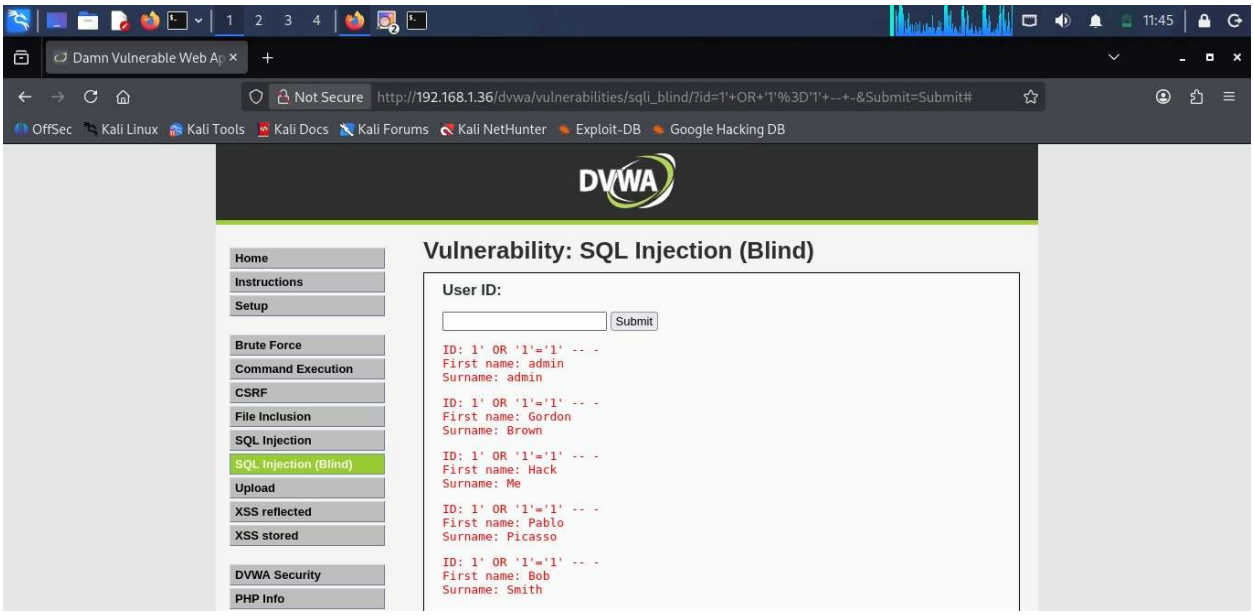

Figure 9: SQL Injection Login Bypass

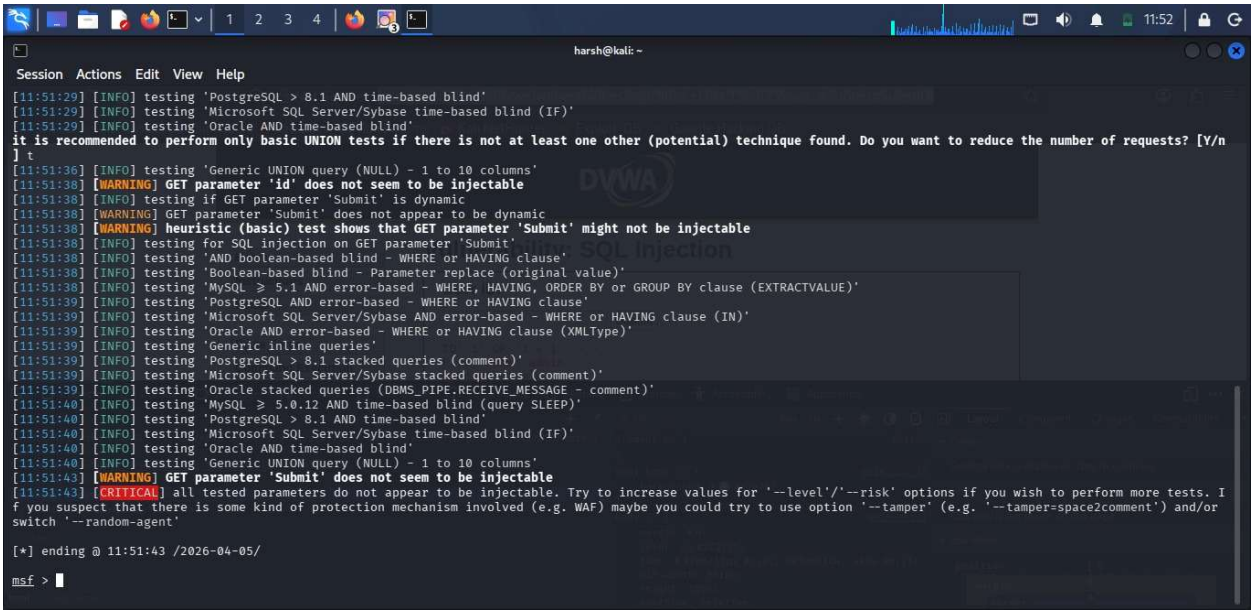

Figure 10: SQLMap Automated Exploitation

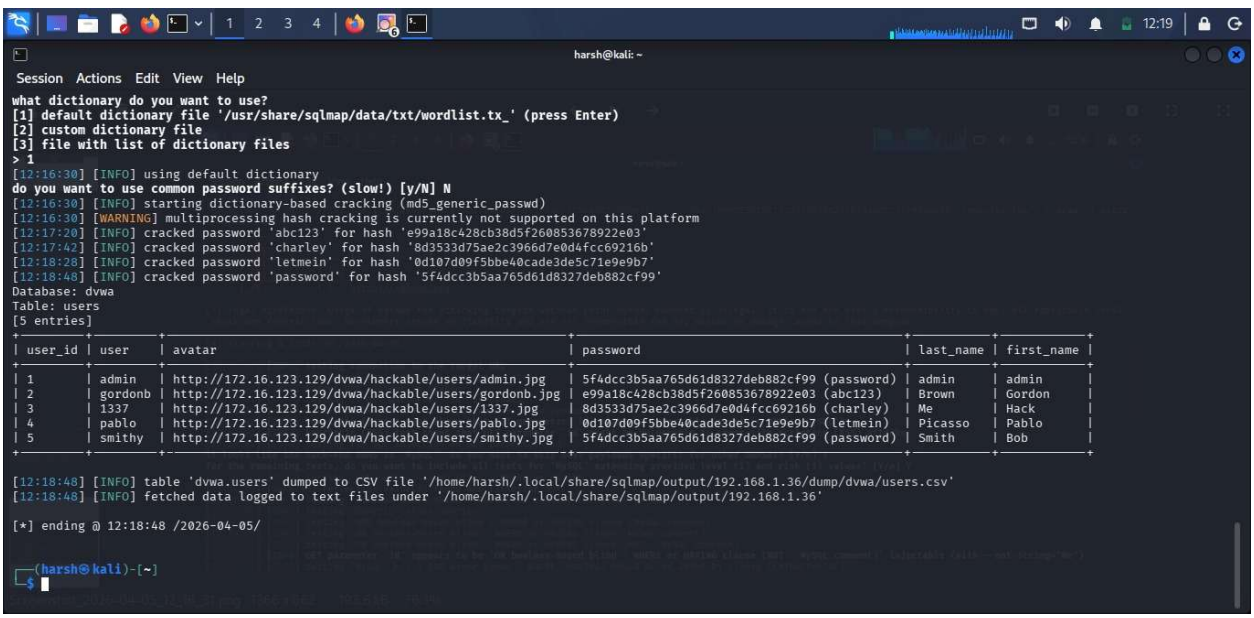

Figure 11: SQLMap Database Extraction

3.2 Cross-Site Scripting (XSS) Testing

XSS vulnerabilities were tested by injecting malicious scripts into input fields. Reflected XSS was identified in the application.

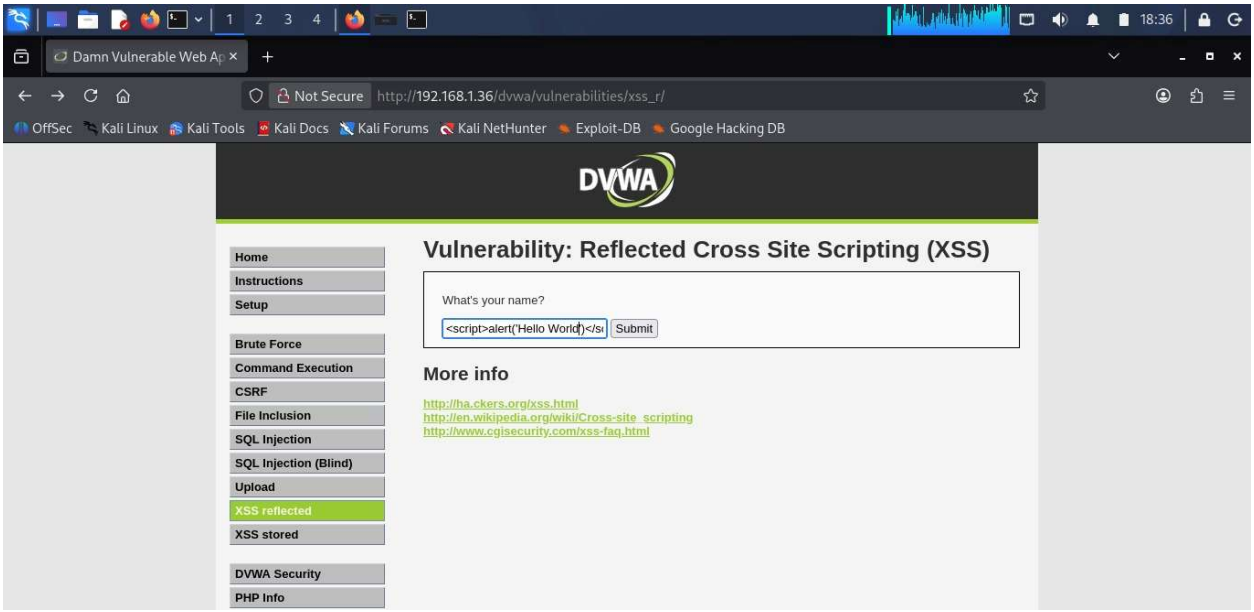

Figure 12: XSS Payload Input

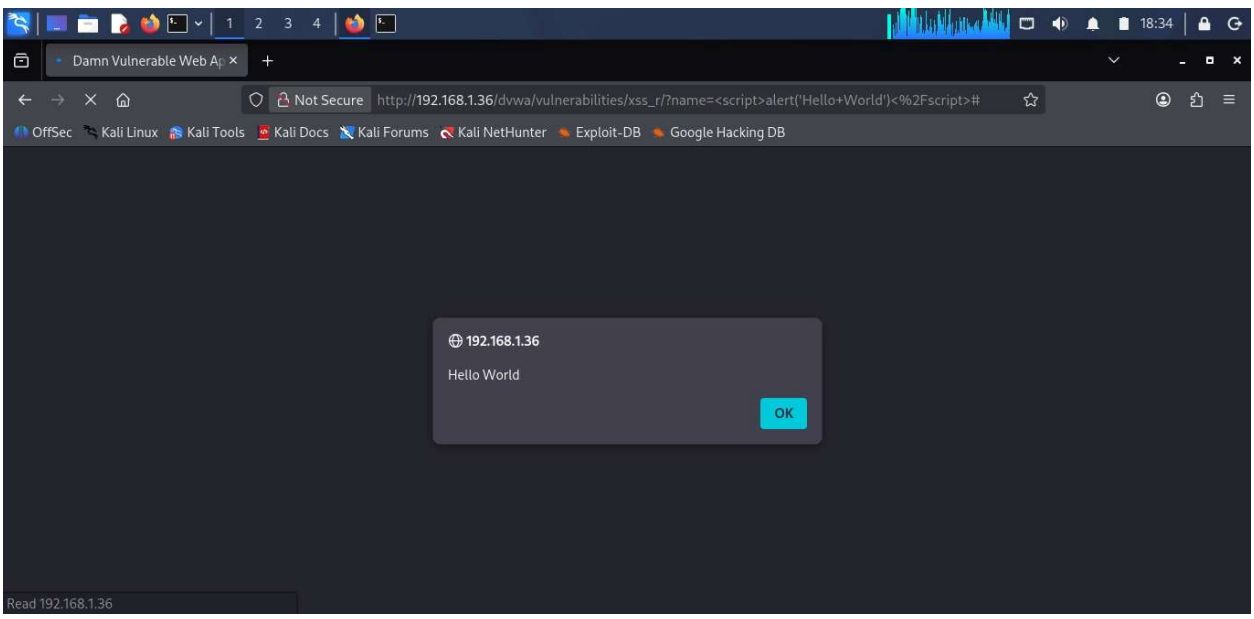

Figure 13: XSS Alert Execution

3.3 Burp Suite Analysis

Burp Suite was used to intercept HTTP requests and analyze parameters. Request manipulation helped identify vulnerabilities.

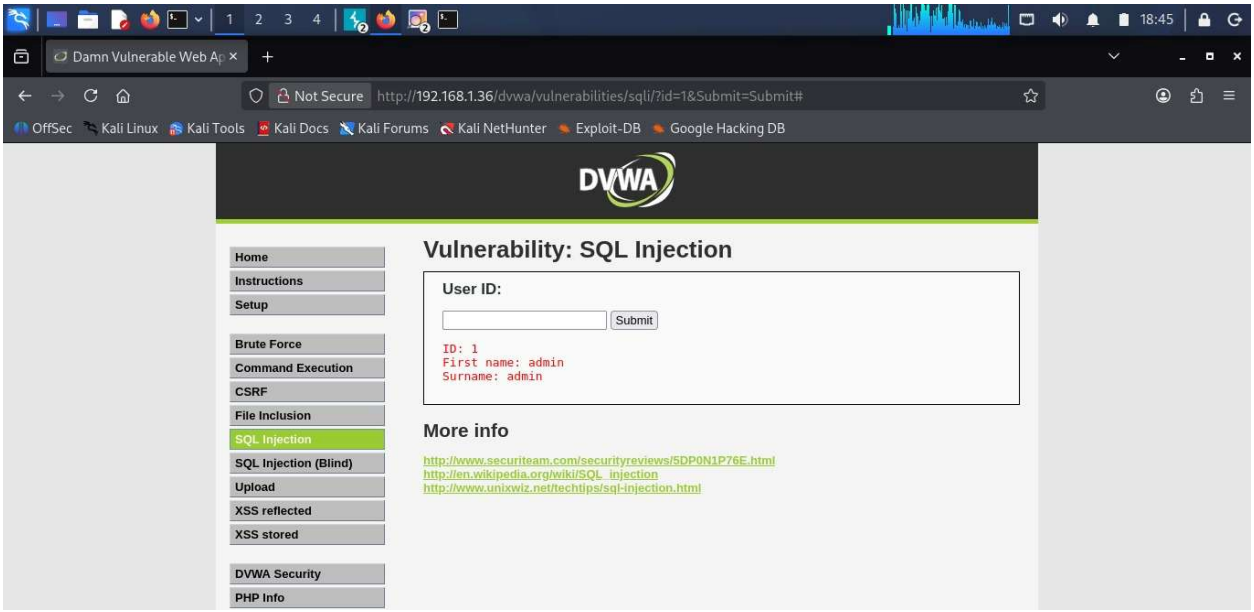

Figure 14: Burp Suite Proxy Interception

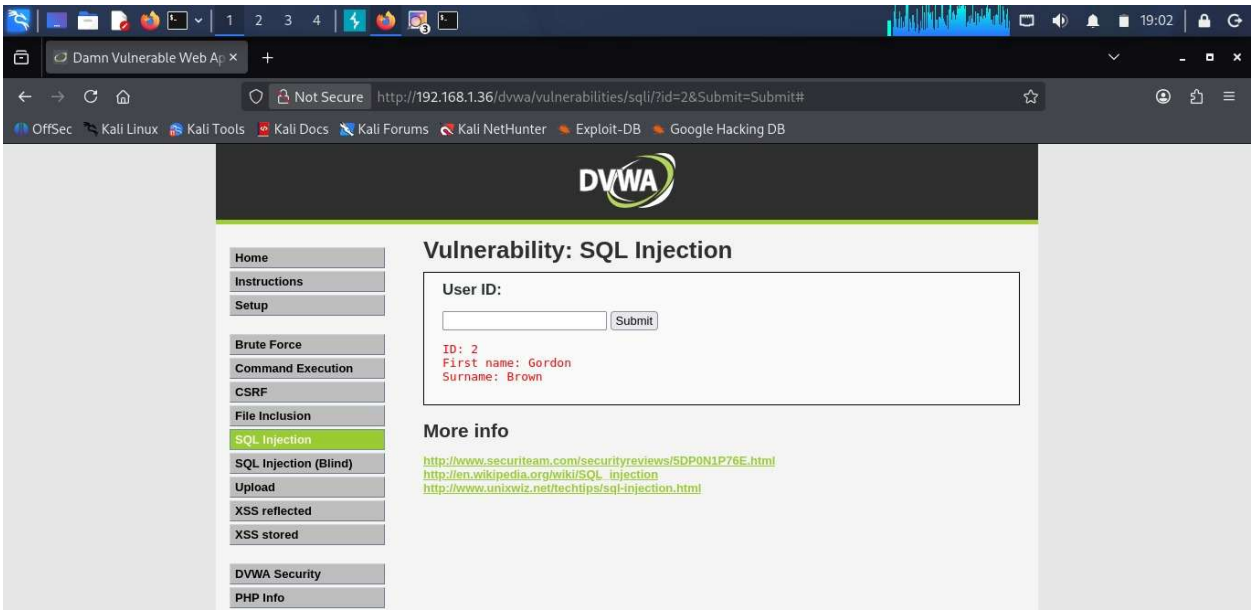

Figure 15: Burp Request Modification

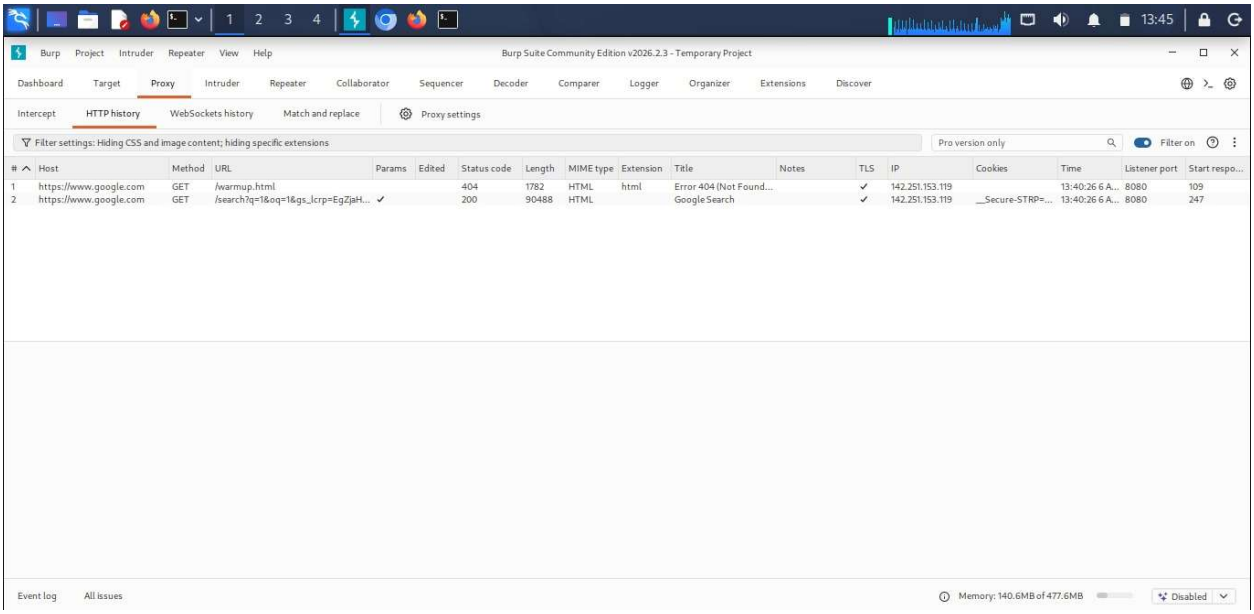

Figure 16: Burp Response Analysis

3.4 Vulnerability Documentation

Test ID	Vulnerability	Severity	Target
001	SQL Injection	Critical	Login Page
002	XSS	Medium	Input Form

Figure 17: Findings Table

4. REPORTING PRACTICE

4.1 Findings Table

Finding ID	Vulnerability	CVSS Score	Remediation
F001	SQL Injection	9.1	Input validation
F002	Weak Password	7.5	Enforce complexity

Figure 18: Findings Table

4.2 Attack Path Diagram

A visual representation of the attack flow was created to demonstrate how vulnerabilities were exploited step-by-step.

Finding ID	Vulnerability	Severity
F001	SQL Injection	Critical
Description	Allows Database Access	
F002	Cross-Site Scripting	Medium
Description	Executes malicious scripts	
F003	Weak Authentication	High
Description	Login bypass possible	

Figure 19: Vulnerability Findings Table

Low	Medium	High	Critical
0.0 - 3.9	4.0 - 6.9	7.0 - 8.9	9.0 - 10

Figure 20: CVSS Risk Assessment

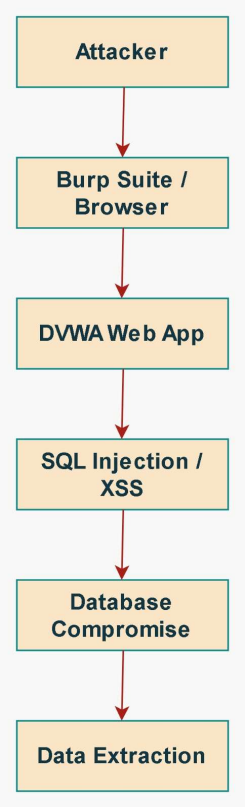

Figure 21: Attack Path Diagram

4.3 Exploit Development Process

The report includes:

- Executive Summary
- Technical Findings
- Risk Assessment
- Remediation Steps

5. POST-EXPLOITATION, EVIDENCE COLLECTION

5.1 Post-Exploitation Activities

After successful exploitation, post-exploitation techniques were performed to assess the depth of system compromise and potential impact.

Privilege Escalation

Privilege escalation techniques were used to gain higher-level (administrative/root) access on the compromised system. This allowed complete control over the system and demonstrated the severity of the vulnerabilities exploited.

System Enumeration

Detailed system enumeration was conducted to gather critical information such as:

- User accounts and privileges
- Running processes
- Installed applications
- Network configurations

Tools like Meterpreter were used to automate and simplify the enumeration process.

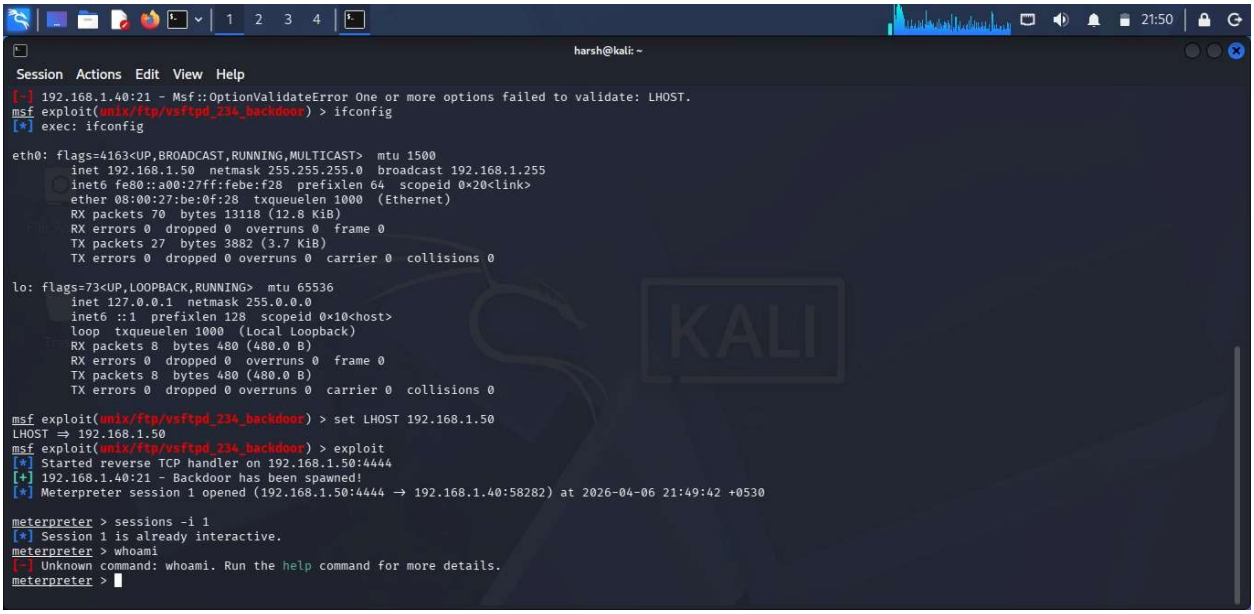

Figure 22: Meterpreter Shell Access

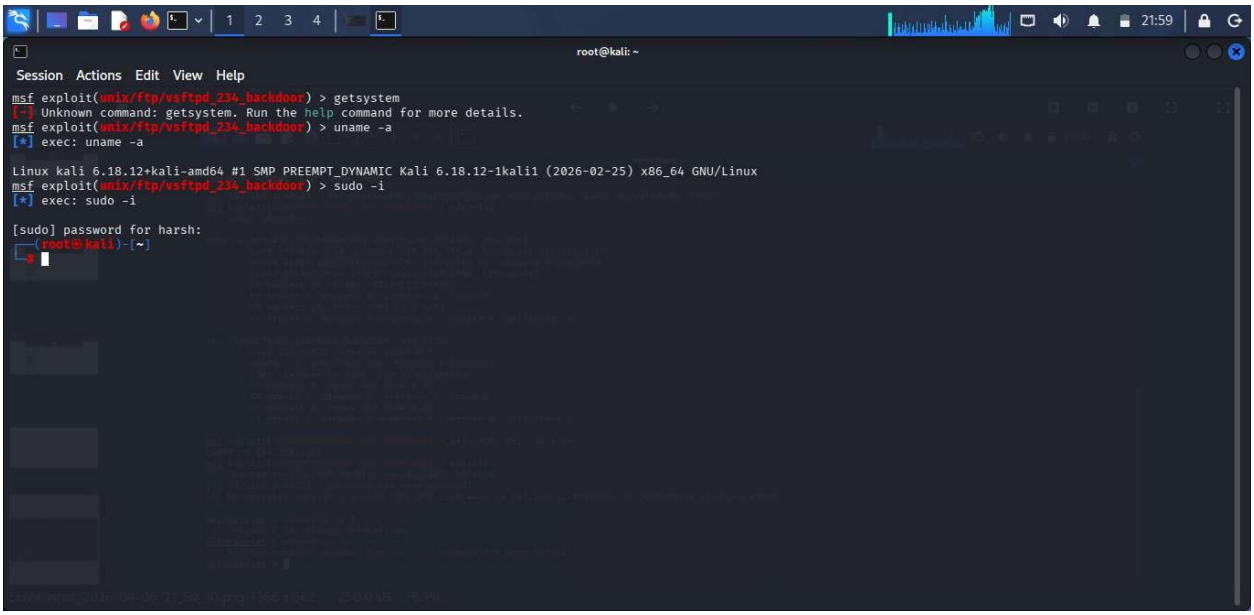

Figure 23: System Access Verification

5.2 Evidence Collection

During the post-exploitation phase, relevant evidence was collected to support findings and ensure proper documentation.

- Network traffic was captured using tools like Wireshark
- System logs and sensitive files were extracted
- All collected data was hashed (e.g., SHA-256) to maintain integrity and authenticity

Proper chain-of-custody procedures were followed to ensure that the evidence remains valid for reporting and analysis purposes.

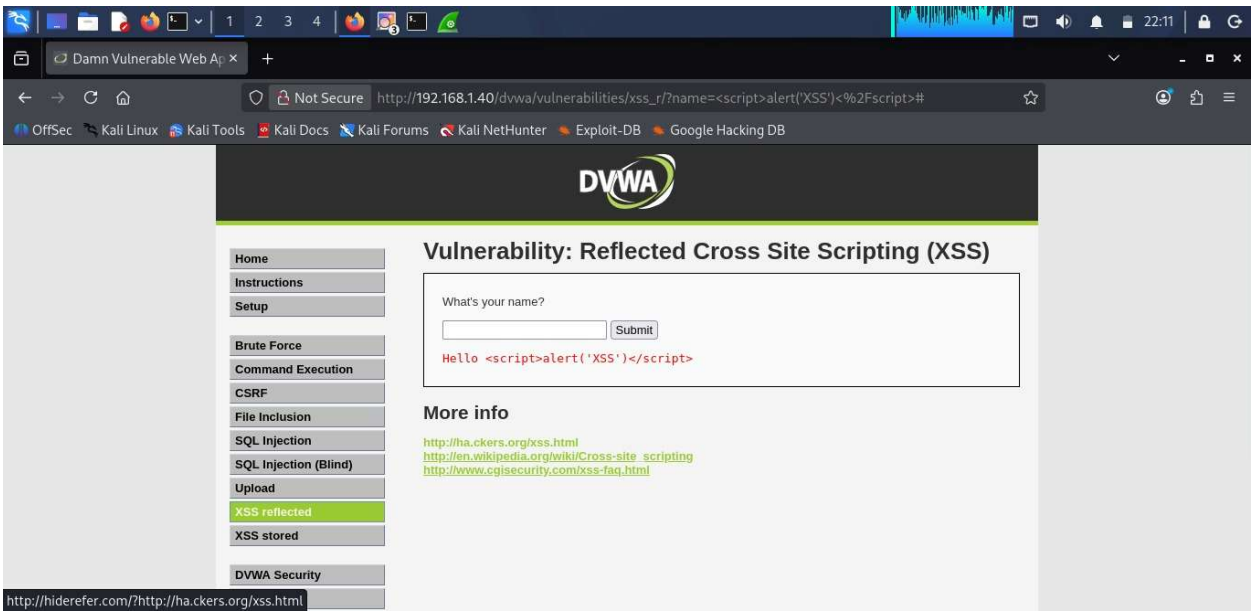

Figure 24: Wireshark Packet Capture

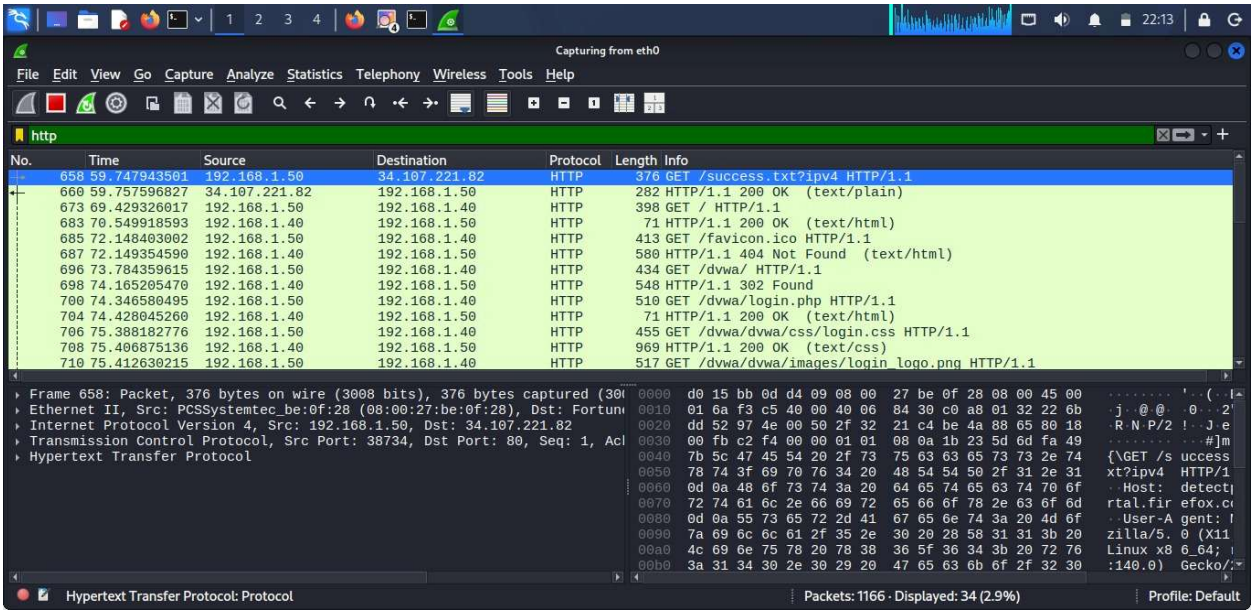

Figure 25: HTTP Traffic Analysis

6. CAPSTONE PROJECT (FULL VAPT CYCLE)

The capstone project involved performing a complete VAPT cycle, simulating a real-world penetration testing engagement.

6.1. Reconnaissance & Scanning

Initial reconnaissance and scanning were performed using tools such as Nmap and OpenVAS to identify open ports, services, and potential vulnerabilities in the target system.

6.2 Vulnerability Assessment

The identified vulnerabilities were analyzed and categorized based on their severity and impact. High-risk vulnerabilities were prioritized for exploitation.

6.3 Exploitation

Exploitation was carried out using Metasploit and other tools to gain unauthorized access to the system. Successful exploitation demonstrated how attackers could compromise the system.

6.4 Vulnerability Assessment

Further actions were performed after gaining access, including privilege escalation, data extraction, and maintaining persistence to simulate real attack scenarios.

6.5 Final Report Summary

The assessment revealed multiple vulnerabilities that could lead to serious security breaches. Key recommendations include:

- Applying security patches regularly
- Implementing strong input validation
- Enhancing access control mechanisms
- Conducting periodic security assessments

7. CONCLUSION AND REFERENCE

7.1. Conclusion

This practical implementation of VAPT provided hands-on experience in identifying and exploiting vulnerabilities. The use of tools like Metasploit and Burp Suite demonstrated real-world attack scenarios. Proper remediation and secure coding practices are essential to protect systems from such threats.

7.2 References

- OWASP Top 10
- Exploit-DB
- Metasploit Documentation
- PortSwigger Web Security Academy
- PTES Framework